

Exam 2P

Course Title: **Terminología Especializada en Tecnología Inglés-Español**
Course ID: **IT0627 (Marron), 26-1**
Cohort ID: **6.2**
Exam Date: **11 Nov 2025**

General Instructions: Print your name in the upper right-hand corner of this paper. Read each item carefully. Be sure that you understand exactly what is being asked of you. Begin your answers on the backside of this paper if possible and add extra sheets of paper as needed. Be sure to write your name on any and all extra sheets of paper. Staple all exam papers together when you are finished.

Do not look at other student's exams. If you have a question or a request during the exam, raise your hand and the instructor will call on you.

This is a closed book exam. Attempt to answer all questions, even if you are uncertain. Whenever possible, provide answers in bullet list format with complete content. Tasks will be evaluated by sub-tasks. Three (3) points are available for each sub-task: Accuracy (1 pt), Completeness (1 pt), and Sufficiency (1 pt). Points will be awarded in 0.1 increments. Answer in English unless requested to do so otherwise.

Task 1 (6 pts)

Science and engineering are often considered to be different professions, and yet both may share a public trust: to be morally and ethically responsible for their work.

1. What is your perspective on the difference and similarity between science and engineering?
2. According to Resnk, what are the three ethical challenges of socially responsible science?
3. Do you think scientists and engineers have a moral or ethical responsibility for the knowledge they create?

Task 2 (6 pts)

Briefly explain the concept of 'wicked problems' and how such problems differ from the problems of 'normal science'.

Task 3 (6 pts)

Translate the following into English:

Los proyectos estratégicos que ha implementado el Tecnológico Nacional de México en el área de semiconductores han sido el diseño y la creación de programas educativos basados en el análisis de la cadena de valor de la industria electrónica partiendo de los materiales semiconductores y caracterización de dispositivos, hasta el diseño y manufactura de los sistemas embebidos o productos electrónicos de consumo, pasando por las etapas del estudio, análisis y caracterización de materiales y dispositivos electrónicos, diseño de circuitos integrados, layout o diseño físico, verificación y empaque de circuitos integrados. Además, se tomaron en cuenta los requerimientos de los principales actores de la industria, como lo son los centros de investigación y las grandes empresas tecnológicas. Se han considerado de manera significativa no solo las áreas de desarrollo, sino esquemas de

capacitación para los profesores de dichos programas educativos. Esto le da gran fortaleza y muy altas posibilidades de éxito a estos nuevos programas educativos.

Task 4 (6 pts)

Answer the following questions:

1. What is a Standard Operating Procedure (SOP)?
2. Why are SOPs written? What is their purpose?
3. What is QA/QC?

Task 5 (6 pts)

Translate the following into Spanish:

A Standard Operating Procedure (SOP) is a set of written instructions that document a routine or repetitive activity followed by an organization. The development and use of SOPs are an integral part of a successful quality system as it provides individuals with the information to perform a job properly, and facilitates consistency in the quality and integrity of a product or end-result. SOPs are intended to be specific to the organization or facility whose activities are described and assist that organization to maintain their quality control and quality assurance processes and ensure compliance with governmental regulations.

Task 6 (6pts)

1. Provide definitions for the following words from Steffen et al. (2018) - Trajectories of the Earth System in the Anthropocene (01...pdf)

Anthropocene
anthropogenic
bifurcation
decarbonization
self-reinforcing feedback loop
oscillations
tipping point

2. Provide definitions for the following words from Poggio et al. (2025) - The Great Wave: Evidence of a large-scale vertical corrugation propagating outwards in the Galactic disc (02....pdf)

accretion
heliocentric
Occam's razor
parsec

Task 7 (6pts)

3. Provide definitions for the following words from Paniagua and Elvira (2010) -Deslizamiento de un talud en el Poniente de la ciudad de México (03....pdf)

anclaje
arena pumítica
estratigráfico

geotécnica
pluvial

4. Provide definitions for the following words from Tao et al. (2025) - Ultrabroadband on-chip photonics for full-spectrum wireless communications (09....pdf)

amplitude
bit
byte
calibration
electromagnetic
gigabyte (Gb)
photon
schematic
spectrum
synergistic
ubiquitous